

Assignment on Friction Problems

Resolving the forces vertically,

$$R_2 = 2 + R_1 = 2 + 0.85 = 2.85 \text{ kN}$$

$\therefore$

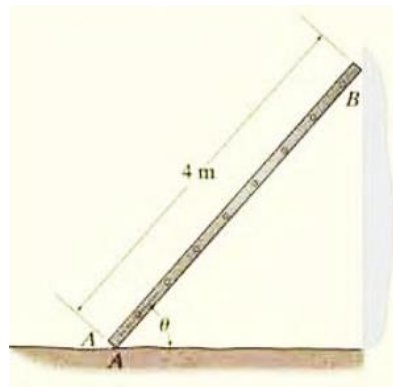
$$F_2 = \mu R_2 = 0.3 \times 2.85 = 0.855 \text{ kN}$$

...(iv)

and now resolving the forces horizontally,

$$P = F_1 + F_2 = 0.255 + 0.855 = 1.11 \text{ kN} \quad \text{Ans.}$$

3. A uniform 10 kg ladder rests against the smooth wall at B and the end A rests on the rough horizontal plane for which the coefficient of static friction is 0.3. Determine the angle of inclination of the ladder and the normal reaction at B if the ladder is on the verge of slipping.



Equations of Equilibrium and Friction. Since the ladder is on the verge of slipping, then $F_A = \mu_s N_A = 0.3N_A$. By inspection, N_A can be obtained directly.

$$+\uparrow \Sigma F_y = 0; \quad N_A - 10(9.81) \text{ N} = 0 \quad N_A = 98.1 \text{ N}$$

Using this result, $F_A = 0.3(98.1 \text{ N}) = 29.43 \text{ N}$. Now N_B can be found.

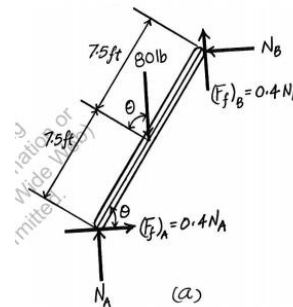
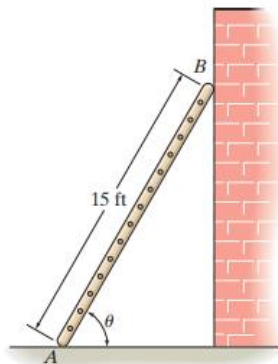
$$\begin{aligned} \rightarrow \Sigma F_x = 0; \quad 29.43 \text{ N} - N_B &= 0 \\ N_B &= 29.43 \text{ N} = 29.4 \text{ N} \quad \text{Ans.} \end{aligned}$$

Finally, the angle θ can be determined by summing moments about point A.

$$\begin{aligned} \zeta + \Sigma M_A = 0; \quad (29.43 \text{ N})(4 \text{ m}) \sin \theta - [10(9.81) \text{ N}](2 \text{ m}) \cos \theta &= 0 \\ \frac{\sin \theta}{\cos \theta} = \tan \theta &= 1.6667 \\ \theta &= 59.04^\circ = 59.0^\circ \quad \text{Ans.} \end{aligned}$$

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4. The ladder has a uniform weight of 80 lb and rests against the wall at B . If the coefficient of static friction at A and B is $\mu = 0.4$, determine the smallest angle at which the ladder will not slip.



SOLUTION

Free-Body Diagram: Since the ladder is required to be on the verge to slide down, the frictional force at A and B must act to the right and upward respectively and their magnitude can be computed using friction formula as indicated on the FBD, Fig. a .

$$(F_f)_A = \mu N_A = 0.4 N_A \quad (F_f)_B = \mu N_B = 0.4 N_B$$

Equations of Equilibrium: Referring to Fig. a .

$$\pm \rightarrow \Sigma F_x = 0; \quad 0.4N_A - N_B = 0 \quad N_B = 0.4 N_A \quad (1)$$

$$+ \uparrow \Sigma F_y = 0; \quad N_A + 0.4N_B - 80 = 0 \quad (2)$$

Solving Eqs. (1) and (2) yields

$$N_A = 68.97 \text{ lb} \quad N_B = 27.59 \text{ lb}$$

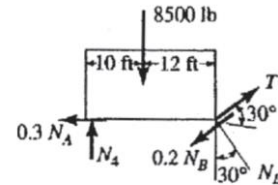
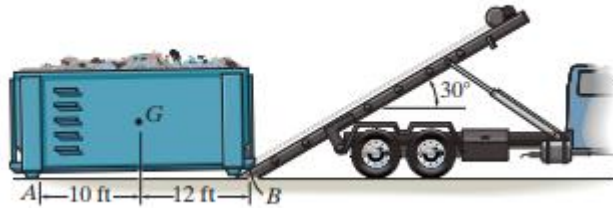
Using these results,

$$\begin{aligned} \zeta + \Sigma M_A &= 0; & 0.4(27.59)(15 \cos \theta) + 27.59(15 \sin \theta) - 80 \cos \theta(7.5) &= 0 \\ & & 413.79 \sin \theta - 434.48 \cos \theta &= 0 \\ \tan \theta &= \frac{\sin \theta}{\cos \theta} = \frac{434.48}{413.79} = 1.05 \\ \theta &= 46.4^\circ \end{aligned}$$

Ans.

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5. The winch on the truck is used to hoist the garbage bin onto the bed of the truck. If the loaded bin has a weight of 8500 lb and center of gravity at G , determine the force in the cable needed to begin the lift. The coefficients of static friction at A and B are $\mu_A = 0.3$ and $\mu_B = 0.2$, respectively. Neglect the height of the support at A .



Hints

SOLUTION

$$\zeta + \sum M_B = 0; \quad 8500(12) - N_A(22) = 0$$

$$N_A = 4636.364 \text{ lb}$$

$$\rightarrow \sum F_x = 0; \quad T \cos 30^\circ$$

$$- 0.2 N_B \cos 30^\circ - N_B \sin 30^\circ - 0.3(4636.364) = 0$$

$$T(0.86603) - 0.67321 N_B = 1390.91$$

$$+\uparrow \sum F_y = 0; \quad 4636.364 - 8500 + T \sin 30^\circ + N_B \cos 30^\circ$$

$$- 0.2 N_B \sin 30^\circ = 0$$

$$T(0.5) + 0.766025 N_B = 3863.636$$

Solving:

$$T = 3666.5 \text{ lb} = 3.67 \text{ kip}$$

$$N_B = 2650.6 \text{ lb}$$

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